Lesson 9.1 - Z-Scores, Confidence Intervals, and Margin of Error

> **Video:** How To Find The Z Score, Confidence Interval, and Margin of Error for a Population Mean by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Confidence interval for μ (when σ is known)

$$\bar{x} - z\,\frac{\sigma}{\sqrt{n}} \;<\; \mu \;<\; \bar{x} + z\,\frac{\sigma}{\sqrt{n}} \qquad \text{or} \qquad \bar{x} \pm \text{EBM}$$

Margin of error, also called the **error bound for the mean (EBM)**:

$$\text{EBM} = z\,\frac{\sigma}{\sqrt{n}}$$

Finding the critical z-value

The area to the **left** of the positive critical value z is

$$\text{left area} = \frac{\text{CL} + 1}{2}$$

where CL is the confidence level as a decimal. Look this area up in a z-table that gives **left** areas.

Confidence level	Left area	z
90%	0.95	1.645
95%	0.975	1.96

For 90%, the table has 0.94950 at $z = 1.64$ and 0.95053 at $z = 1.65$. 0.95 is about halfway between them, so average: $z = 1.645$.

Example 1 - physics test, 95% CI

$\sigma = 5.4$, $n = 50$, $\bar{x} = 79$

1. z: left area $= (0.95 + 1)/2 = 0.975 \rightarrow z = 1.96$
2. Standard error: $\dfrac{5.4}{\sqrt{50}} \approx 0.76368$
3. **(b) Margin of error:** $\text{EBM} = 1.96 \times 0.76368 \approx 1.4968$
4. **(a) Interval:** 79 ± 1.4968

$$77.5032 < \mu < 80.4968$$

We are 95% confident that the mean score of the whole class lies between about **77.50 and 80.50**.

Example 2 - chemistry test, 90% CI

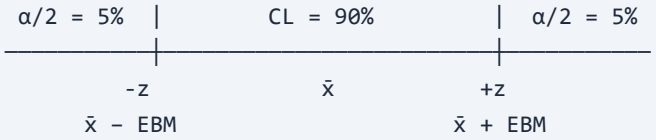
$\sigma = 6.5$, $n = 100$, $\bar{x} = 82$

1. z: left area $= (0.90 + 1)/2 = 0.95 \rightarrow z = 1.645$
2. **(b) Margin of error:** $\text{EBM} = 1.645 \times \dfrac{6.5}{\sqrt{100}} = 1.645 \times 0.65 = 1.06925$
3. **(a) Interval:** 82 ± 1.06925

$$80.93075 < \mu < 83.06925$$

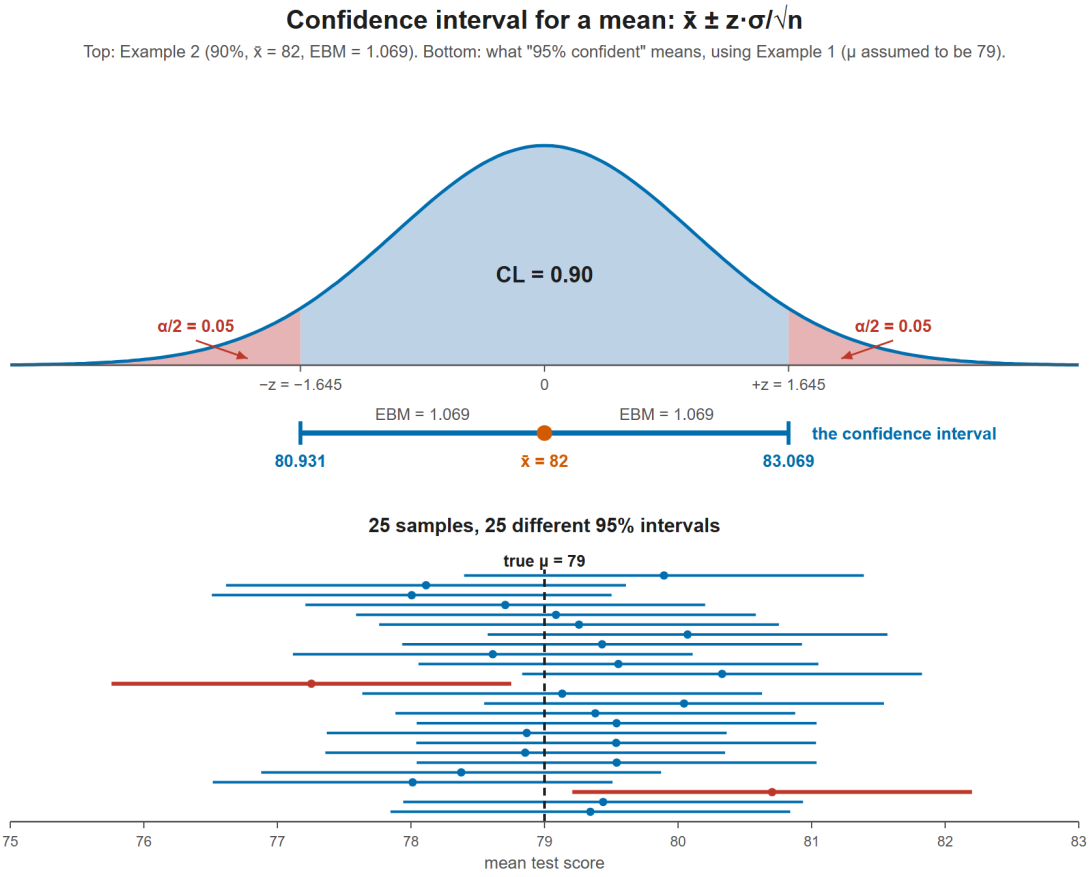
Confidence interval vs confidence level

Term	Meaning	Example
Confidence interval (CI)	The **range of values** $(\bar{x} - \text{EBM},\ \bar{x} + \text{EBM})$	80.93 to 83.07
Confidence level (CL)	The **area under the curve** between $-z$ and $+z$	0.90
α	The area **outside** the interval: $\alpha = 1 - \text{CL}$, split as $\alpha/2$ in each tail	0.10 → 0.05 per tail



Interpreting "95% confidence"

If we took many samples and built a 95% interval from each one, about **95% of those intervals would contain the true mean** μ, and about 5% would miss it. The video phrases this in terms of sample means landing in the interval; this is the more precise statement.



In this run 2 of the 25 intervals (red) miss μ. In the long run about 5% of 95% intervals miss.

Top: the confidence level is the middle area between â□□z and +z, and the interval reaches one EBM either side of xì□. Bottom: 25 simulated samples give 25 different intervals. Most of them cover the true mean (dashed line), but a few miss it, which is what "95% confident" means in the long run.

What changes the margin of error?

$$\text{EBM} = z \frac{\sigma}{\sqrt{n}}$$

Change	Effect on EBM	Why
Higher confidence level (90% → 95%)	**Larger**	z grows, so the interval gets wider
Larger sample size n	**Smaller**	$\sqrt{n}$ is in the denominator

To get a smaller margin of error, **lower the confidence level** or **increase the sample size**.

The EBM is half the width of the confidence interval.

Key takeaways

- CI = $\bar{x} \pm z\,\sigma/\sqrt{n}$, and EBM = $z\,\sigma/\sqrt{n}$.
- Critical z: look up the left area $(\mathrm{CL}+1)/2$. 90% → 1.645, 95% → 1.96.
- Higher confidence gives a wider interval. A bigger sample gives a narrower one.

Lesson 9.2 - How to Find the Z-score Given the Confidence Intervals

> **Video:** How To Find The Z Score Given The Confidence Level of a Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Common values to memorize

Confidence level	z
90%	1.645
95%	1.96
98%	2.33
99%	2.575

Method for any confidence level

Step 1 - area to the left of $+z$:

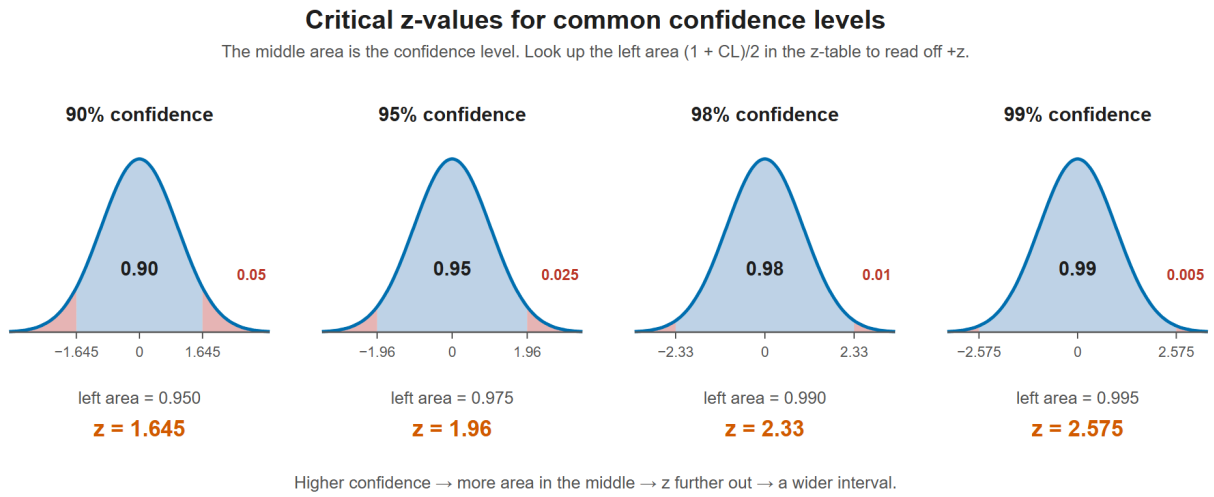
$$\text{left area} = \frac{1 + \text{CL}}{2}$$

Why: the confidence level is the area between $-z$ and $+z$. The leftover $1 - \text{CL}$ is split evenly between the two tails. For 95%:

```
 0.025  |          0.95           |  0.025
────────┼─────────────────────────┼────────
       -z           mean         +z
left area up to +z = 0.95 + 0.025 = 0.975
```

Step 2 - find that area inside a positive z-table (one that gives **left** areas). Read the row (ones and tenths digits) and the column (hundredths digit), then add them.

Example: 0.975 sits in row **1.9**, column **0.06**, so $z = 1.9 + 0.06 = 1.96$.



The confidence level is the middle area, and each tail gets half of what is left. The left area up to +z is (1 + CL)/2, which is the number to look up in the z-table.

Practice

97% confidence

$$\text{left area} = \frac{1 + 0.97}{2} = 0.985 \;\Rightarrow\; \text{row } 2.1,\ \text{column } 0.07 \;\Rightarrow\; z = 2.17$$

92% confidence

$$\text{left area} = \frac{1 + 0.92}{2} = 0.96$$

The closest table entry is 0.95994, in row **1.7**, column **0.05**, so $z = 1.75$.

Key takeaways

- Left area $= (1 + \text{CL})/2$. Look it up in the body of the z-table to read off z.
- If the exact area isn't in the table, use the closest entry, or average two entries that are equally close (e.g. 90% → 1.645).
- Know 1.645, 1.96, 2.33 and 2.575 by heart.

Lesson 9.3 - How to Calculate the Sample Size

> **Video:** How To Calculate The Sample Size Given The Confidence Level & Margin of Error by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Formulas

For a mean. Start from the margin of error:

$$E = z\,\frac{\sigma}{\sqrt{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\sigma^2}{n} \;\Rightarrow\; \boxed{n = \frac{z^2\sigma^2}{E^2}}$$

For a proportion:

$$\boxed{n = \frac{z^2\,\hat{p}\,(1-\hat{p})}{E^2}}$$

If $\hat{p}$ is unknown, use $\hat{p} = 0.5$. That makes $\hat{p}(1-\hat{p})$ as large as possible, so the resulting n is big enough whatever the true proportion is.

Always **round** n **up** to the next whole number.

Example 1 - working backwards to the class size

The mean test score was 84 with standard deviation $\sigma = 15$. The margin of error at 95% confidence was $E = 5.367$. How many students were sampled?

$$n = \frac{(1.96)^2(15)^2}{(5.367)^2} \approx 30.0 \;\Rightarrow\; n = 30 \text{ students}$$

(The mean, 84, isn't needed.)

Example 2 - planning a survey on a proportion

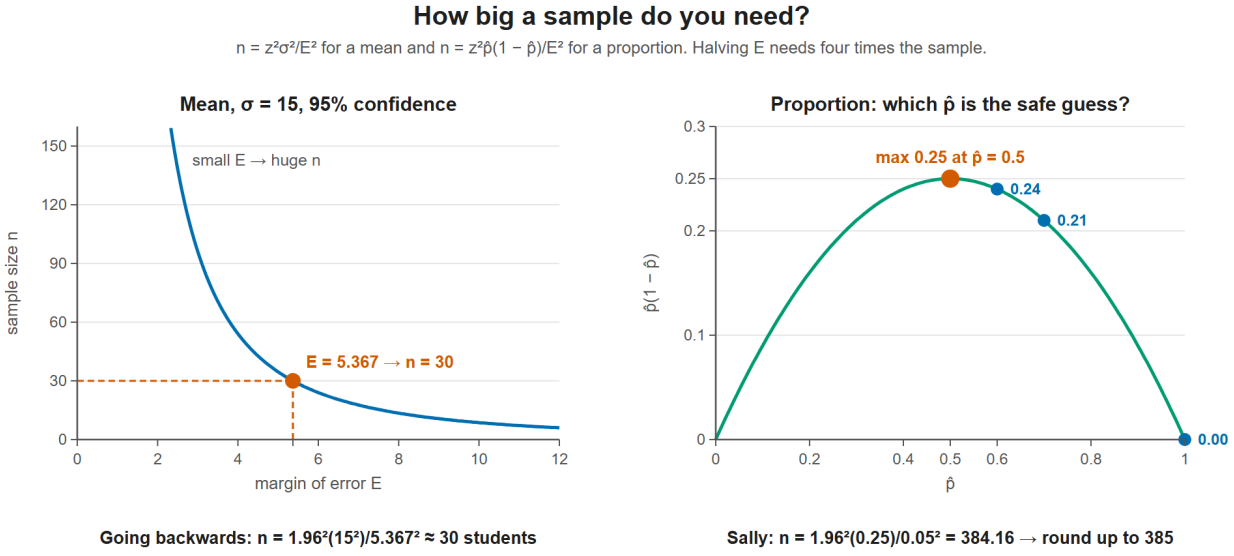
Sally wants to estimate the proportion of high-school students planning to go to college, with **95%** confidence and a **5%** margin of error ($E = 0.05$). She has no prior estimate of $\hat{p}$.

Which $\hat{p}$ makes $\hat{p}(1-\hat{p})$ largest?

$\hat{p}$	$1-\hat{p}$	**Product**
1.0	0.0	0.00
0.7	0.3	0.21
0.6	0.4	0.24
0.5	**0.5**	**0.25** (largest)

$$n = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2} = \frac{0.9604}{0.0025} = 384.16 \;\Rightarrow\; n = 385$$

Round **up**: 384 students would not quite reach the 5% margin.



Left: the required n grows fast as the margin of error shrinks, and Example 1's E = 5.367 corresponds to n = 30. Right: pÌ□(1 â□□ pÌ□) is largest at pÌ□ = 0.5, so using 0.5 when nothing is known gives the safest (largest) sample size.

Key takeaways

- Mean: $n = z^2\sigma^2/E^2$. Proportion: $n = z^2\hat{p}(1-\hat{p})/E^2$.
- With no estimate of $\hat{p}$, use 0.5. This gives the largest, safest n.
- Always round **up**.
- A smaller margin of error or a higher confidence level requires a larger sample.

Lesson 9.4 - Student's T-Distribution

> **Video:** Student's T Distribution - Confidence Intervals & Margin of Error by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

When to use t instead of z

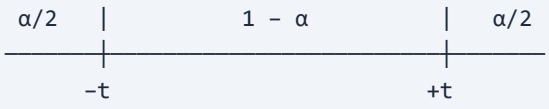
Use the **normal (z)** distribution when...	Use **Student's t** when...
the **population** SD σ is known	only the **sample** SD s is known
$n \geq 30$	$n < 30$

Having only s and a small n are both signs that you need t.

Formulas

$$\bar{x} \pm t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}}, \qquad \text{EBM} = t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}}$$

Symbol	Meaning
$\nu = n - 1$	**Degrees of freedom** (Greek "nu")
$\alpha = 1 - \text{CL}$	The total area outside the interval, i.e. both tails
$\alpha/2 = \dfrac{1 - \text{CL}}{2}$	The **one-tail** area, used to look up t



Reading a t-table: go to the row for ν and the column for the one-tail area $\alpha/2$.

Example 1 - student weights, 95% CI

20 students: $\bar{x} = 165$ lb, $s = 4.5$ lb.

s is a **sample** SD (the school has far more than 20 students) and $n = 20 < 30$, so use t.

- $\nu = 20 - 1 = 19$
- $\alpha/2 = (1 - 0.95)/2 = 0.025$
- Table: $t_{19,\,0.025} = 2.093$

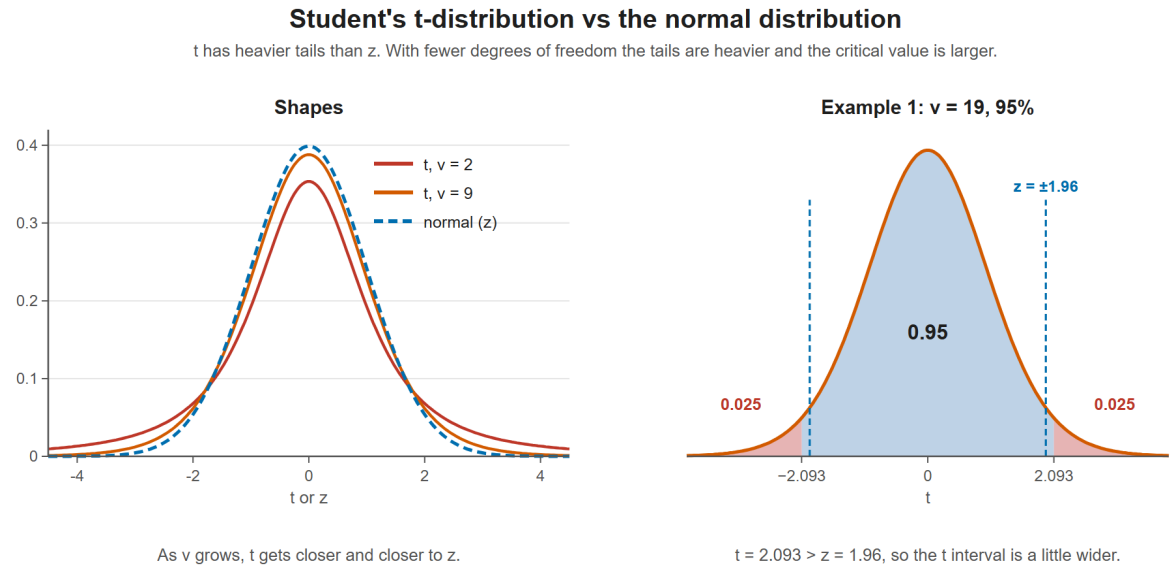
(b) Margin of error

$$\text{EBM} = 2.093 \times \frac{4.5}{\sqrt{20}} = \frac{9.4185}{4.4721} \approx 2.106$$

(a) Interval

$$165 \pm 2.106 \;\Rightarrow\; 162.894 < \mu < 167.106$$

We are 95% confident that the mean weight of all the school's students is between about **162.9 and 167.1 lb**.



Left: t has heavier tails than the normal curve, especially with few degrees of freedom. Right: for Î½ = 19 the 95% critical value is 2.093, a bit further out than z = 1.96, so t intervals are slightly wider.

Example 2 - chemistry scores, 90% CI

The class has 500 students. 10 scores were sampled: 76, 84, 69, 92, 58, 89, 73, 97, 85, 77.

(a) Sample mean and SD

$$\bar{x} = \frac{800}{10} = 80$$

Distances from 80: 4, 4, 11, 12, 22, 9, 7, 17, 5, 3 (squaring removes the signs).

$$s = \sqrt{\frac{4^2 + 4^2 + 11^2 + 12^2 + 22^2 + 9^2 + 7^2 + 17^2 + 5^2 + 3^2}{10 - 1}} = \sqrt{\frac{1234}{9}} \approx 11.709$$

(b) Margin of error at 90%

- $\nu = 10 - 1 = 9$
- $\alpha/2 = (1 - 0.90)/2 = 0.05$

- Table: $t_{9,\,0.05} = 1.8331$

$$\text{EBM} = 1.8331 \times \frac{11.709}{\sqrt{10}} \approx 6.787$$

(c) 90% confidence interval

$$80 \pm 6.787 \;\Rightarrow\; 73.213 < \mu < 86.787$$

Key takeaways

- Use t when σ is unknown (you only have s), especially when $n < 30$.
- Degrees of freedom $\nu = n - 1$. Look up the one-tail area $\alpha/2 = (1 - \text{CL})/2$.
- CI = $\bar{x} \pm t\, s/\sqrt{n}$. The t value is larger than z for small samples, which gives a wider and more cautious interval.

Lesson 9.5 - Finding a Confidence Interval of a Population Proportion

> **Video:** Finding The Confidence Interval of a Population Proportion Using The Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Formulas

Quantity	Formula
Sample proportion ("p-hat" or "p-prime")	$\hat{p} = \frac{x}{n}$, where x = number of "yes" (successes)
Proportion of failures	$\hat{q} = 1 - \hat{p}$
Normal approximation is OK if	$n\hat{p}\hat{q} \geq 10$ (many texts check $n\hat{p} \geq 10$ and $n\hat{q} \geq 10$)
Error bound for a proportion (EBP)	$E = z\sqrt{\frac{\hat{p}\hat{q}}{n}}$
Confidence interval	$\hat{p} - E < p < \hat{p} + E$
Sample size needed	$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2}$, rounded **up**
Critical z	look up the left area $\frac{1 + \mathrm{CL}}{2}$

Common z values: 90% → 1.645, 95% → 1.96, 98% → 2.33, 99% → 2.575.

Example 1 - laptop ownership

800 students were surveyed and 584 own a laptop.

(a) Can the normal distribution be used? This is a binomial situation (yes/no answers).

$$\hat{p} = \frac{584}{800} = 0.73, \quad \hat{q} = 0.27, \quad n\hat{p}\hat{q} = 800(0.73)(0.27) = 157.68 \geq 10 \checkmark$$

(b) 95% confidence interval and margin of error ($z = 1.96$, since the left area is 0.975)

$$E = 1.96\sqrt{\frac{0.73 \times 0.27}{800}} = 1.96\sqrt{2.46375 \times 10^{-4}} \approx 0.0308$$

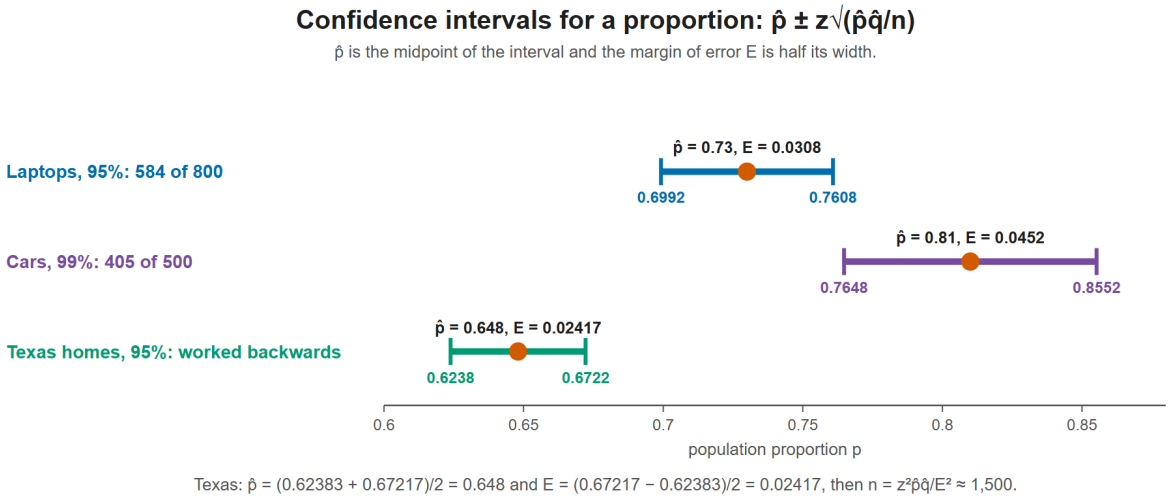
$$0.73 - 0.0308 < p < 0.73 + 0.0308 \;\Rightarrow\; 0.6992 < p < 0.7608$$

We are 95% confident that between about **69.9% and 76.1%** of all students own a laptop.

(c) Sample size for a 2% margin of error at 90% confidence ($z = 1.645$)

$$n = \frac{(1.645)^2(0.73)(0.27)}{(0.02)^2} \approx 1333.39 \;\Rightarrow\; n = 1334$$

Derivation: square $E = z\sqrt{\hat{p}\hat{q}/n}$ to get $E^2 = z^2\hat{p}\hat{q}/n$, then solve for n.



Texas: p̂ = (0.62383 + 0.67217)/2 = 0.648 and E = (0.67217 − 0.62383)/2 = 0.02417, then n = z²p̂q̂/E² ≈ 1,500.

The three intervals from the lesson. The dot is pÌ□, the midpoint, and each arm is the margin of error E. The 99% car interval is the widest, because higher confidence uses a larger z.

Example 2 - car ownership

500 residents were surveyed and 405 own a car.

$$\hat{p} = \frac{405}{500} = 0.81, \qquad \hat{q} = 0.19$$

(a) 99% confidence interval ($z = 2.575$, since the left area 0.995 lies between $z = 2.57$ and 2.58)

$$E = 2.575\sqrt{\frac{0.81 \times 0.19}{500}} = 2.575(0.01754) \approx 0.0452$$

$$0.7648 < p < 0.8552$$

(b) Sample size for a 3% margin of error at 98% confidence ($z = 2.33$, left area 0.99)

$$n = \frac{(2.33)^2(0.81)(0.19)}{(0.03)^2} \approx 928.34 \;\Rightarrow\; n = 929$$

Example 3 - working backwards from an interval

A 95% confidence interval for the proportion of homeowners in Texas is $(0.62383,\ 0.67217)$.

(a) Sample proportion. It is the **midpoint** of the interval:

$$\hat{p} = \frac{0.62383 + 0.67217}{2} = 0.648$$

(b) Error bound. It is **half the width** of the interval:

$$E = \frac{0.67217 - 0.62383}{2} = 0.02417 \quad (= 0.67217 - 0.648)$$

(c) How many people were surveyed?

$$n = \frac{(1.96)^2(0.648)(0.352)}{(0.02417)^2} \approx 1499.95 \;\Rightarrow\; n \approx 1500$$

(d) How many said yes?

$$x = n\hat{p} = 1500(0.648) = 972$$

Key takeaways

- $\hat{p} = x/n$, $\hat{q} = 1 - \hat{p}$. Check $n\hat{p}\hat{q} \geq 10$ before using the normal approximation.
- CI = $\hat{p} \pm z\sqrt{\hat{p}\hat{q}/n}$.
- Midpoint of a CI = $\hat{p}$. Half-width = margin of error.
- Sample size: $n = z^2\hat{p}\hat{q}/E^2$, always rounded **up**.

Lesson 9.6 - Chebyshev's Theorem

> **Video:** Chebyshev's Theorem by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Statement

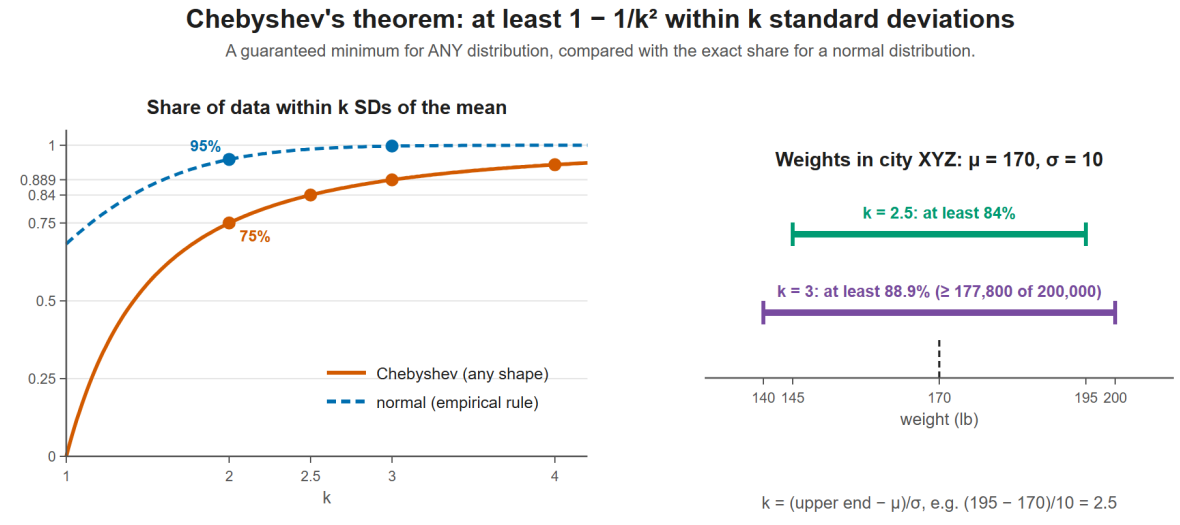
For **any** distribution, whatever its shape, the proportion of data within k standard deviations of the mean is **at least**

$$1 - \frac{1}{k^2}, \qquad k > 1$$

- It gives a **minimum**, not an exact value. The true proportion can be higher.
- It works for **all** distributions, not only normal ones.
- In the video the formula is spoken as "1 over 1 minus k squared". The correct form is $1 - 1/k^2$.

Chebyshev vs the empirical rule

Within	Empirical rule (normal only)	Chebyshev (any distribution, minimum)
1σ	68%	— (k must be > 1)
2σ	95%	$1 - \frac{1}{4} = 75\%$
3σ	99.7%	$1 - \frac{1}{9} \approx 88.9\%$
4σ	≈ 99.99%	$1 - \frac{1}{16} = 93.75\%$



The orange curve is always below the blue one: Chebyshev is a cautious lower bound.

Left: Chebyshev's guaranteed minimum (orange) is always below the exact share for normal data (blue). For example, it guarantees 75% within 2 SDs, while a normal distribution has 95%. Right: the two intervals from the weight example.

Finding k

If an interval is centered on the mean, $\mu \pm k\sigma$, then

$$k = \frac{x - \mu}{\sigma}$$

using the upper end x (the lower end gives the same k).

Example - weights in city XYZ

Mean weight $\mu = 170$ lb, $\sigma = 10$ lb, distribution shape unknown.

(a) Minimum percentage of people weighing between 145 and 195 lb

Both ends are 25 lb from the mean, so

$$k = \frac{195 - 170}{10} = 2.5$$

$$1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = 0.84$$

At least 84% of residents weigh between 145 and 195 lb. The real figure could be 86%, 89%, 95%, ...

(b) The city has 200,000 residents. What is the minimum number weighing between 140 and 200 lb?

Both ends are 30 lb from the mean, so

$$k = \frac{200 - 170}{10} = 3, \qquad 1 - \frac{1}{9} = \frac{8}{9} \approx 0.889$$

$$\frac{8}{9} \times 200{,}000 \approx 177{,}778 \text{ people}$$

(The video rounds to 0.889 first and gets 177,800.)

So $P(140 \le X \le 200) \ge 0.889$, and **at least about 177,800** residents weigh between 140 and 200 lb.

Key takeaways

- Chebyshev: at least $1 - 1/k^2$ of any data set lies within k SDs of the mean ($k > 1$).

- Use it when the distribution is **unknown or not normal**. Use the empirical rule only for normal data.
- $k = (x - \mu)/\sigma$ for an interval centered on the mean.
- Answers are **lower bounds**: say "at least".